CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 8: **Introduction to Trigonometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $\sin\theta = \frac{24}{25}$, the value of $2\sin\theta\cos\theta$ is: 1

(a) $\frac{336}{625}$ **(b)** $\frac{168}{625}$ **(c)** $\frac{14}{25}$ **(d)** $\frac{48}{25}$

2. If $\sec\theta + \tan\theta = 7$, then $\sec\theta - \tan\theta =$ 1

(a) 2 **(b)** 7 **(c)** $\frac{1}{7}$ **(d)** 1

3. $\dfrac{\sin^2 33^\circ + \sin^2 57^\circ}{\cos^2 71^\circ + \cos^2 19^\circ} =$ 1

(a) 1 **(b)** $\frac{1}{2}$ **(c)** 0 **(d)** 2

4. The value of $\tan 33^\circ \tan 57^\circ \tan 33^\circ \tan 57^\circ$ is: 1

(a) 1 **(b)** 2 **(c)** 0 **(d)** $\tan 45^\circ + 1$

5. If $\sin\theta = \frac{21}{29}$ then $\sin^2\theta + \cos^2\theta + \tan^2\theta =$ 1

(a) $\frac{841}{400}$ **(b)** $\frac{441}{841}$ **(c)** $\frac{400}{841}$ **(d)** $\frac{441}{400}$

6. If $\tan\theta = \frac{21}{20}$, then $\dfrac{1-\tan^2\theta}{1+\tan^2\theta} =$ 1

(a) $-\frac{41}{841}$ **(b)** $\frac{840}{841}$ **(c)** $\frac{20}{21}$ **(d)** $\frac{41}{841}$

7. If $\sin\theta = \frac{21}{29}$, the value of $2\sin\theta\cos\theta$ is: 1

(a) $\frac{40}{29}$ **(b)** $\frac{420}{841}$ **(c)** $\frac{42}{29}$ **(d)** $\frac{840}{841}$

8. If $\sec\theta + \tan\theta = \frac{5}{2}$, then $\sec\theta - \tan\theta =$ 1

(a) 2 **(b)** 1 **(c)** $\frac{5}{2}$ **(d)** $\frac{2}{5}$

9. **Assertion (A):** If $\sin A = \dfrac{3}{5}$ for an acute angle A, then $\cos A = \dfrac{4}{5}$. 1

Reason (R): $\sin^2 A + \cos^2 A = 1$ for every angle A.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** $1 + \tan^2\theta = \sec^2\theta$ for every acute angle θ. 1

Reason (R): $1 + \cot^2\theta = \sec^2\theta$ for every acute angle θ.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If $x = a\sec\theta$ and $y = b\tan\theta$ with $a = 4$, $b = 7$, express $\dfrac{x^2}{16} - \dfrac{y^2}{49}$. 2

12. If $\sin\theta = \frac{24}{25}$, find the value of $\cos^2\theta - \sin^2\theta$. 2

13. If $\cot\theta = \frac{12}{35}$, find $\sin\theta$ and $\sec\theta$. 2

14. If $x = a\sec\theta$ and $y = b\tan\theta$ with $a = 5$, $b = 2$, express $\dfrac{x^2}{25} - \dfrac{y^2}{4}$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that: $\dfrac{\sin A}{\csc A} + \dfrac{\cos A}{\sec A} = 1$. 3

16. If $\sin\theta + \cos\theta = \frac{9}{7}$, find $\sin\theta\cos\theta$. 3

17. If $\tan\theta + \cot\theta = 7$, find $\tan^2\theta + \cot^2\theta$. 3

OR

Prove that: $\sin^2 A\cot^2 A + \cos^2 A\tan^2 A = 1$. 3

18. If $\sin\theta + \cos\theta = \frac{11}{8}$, find $\sin\theta\cos\theta$. 3

OR

If $\tan\theta + \cot\theta = 8$, find $\tan^2\theta + \cot^2\theta$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A teacher draws a right triangle for an acute angle θ with $\cot\theta = \dfrac{7}{24}$. 5

(i) Find the hypotenuse if the sides are in the ratio of the given cotangent (take 7k and 24k).

(ii) Find $\sin\theta$ and $\cos\theta$.

(iii) Find $\tan\theta$.

(iv) Evaluate $\dfrac{\sin 35^\circ}{\cos 55^\circ} + \dfrac{\cot 20^\circ}{\tan 70^\circ}$ using complementary angles.

20. Prove that: $\cos^4 A - \sin^4 A = 2\cos^2 A - 1$. 5

OR

If $\tan A = \dfrac{16}{12}$, evaluate $\dfrac{6\sin A + 8\cos A}{1\sin A + 1\cos A}$. 5

END OF QUESTION PAPER — **Total: 40 Marks**

MASTERMIND